MOCKUP AI

TEAM NAME- **NIGHTSHADES**

TEAM LEAD COLLEGE NAME- **SARASWATI COLLEGE OF ENGINEERING**

CITY - **KHARGHAR**

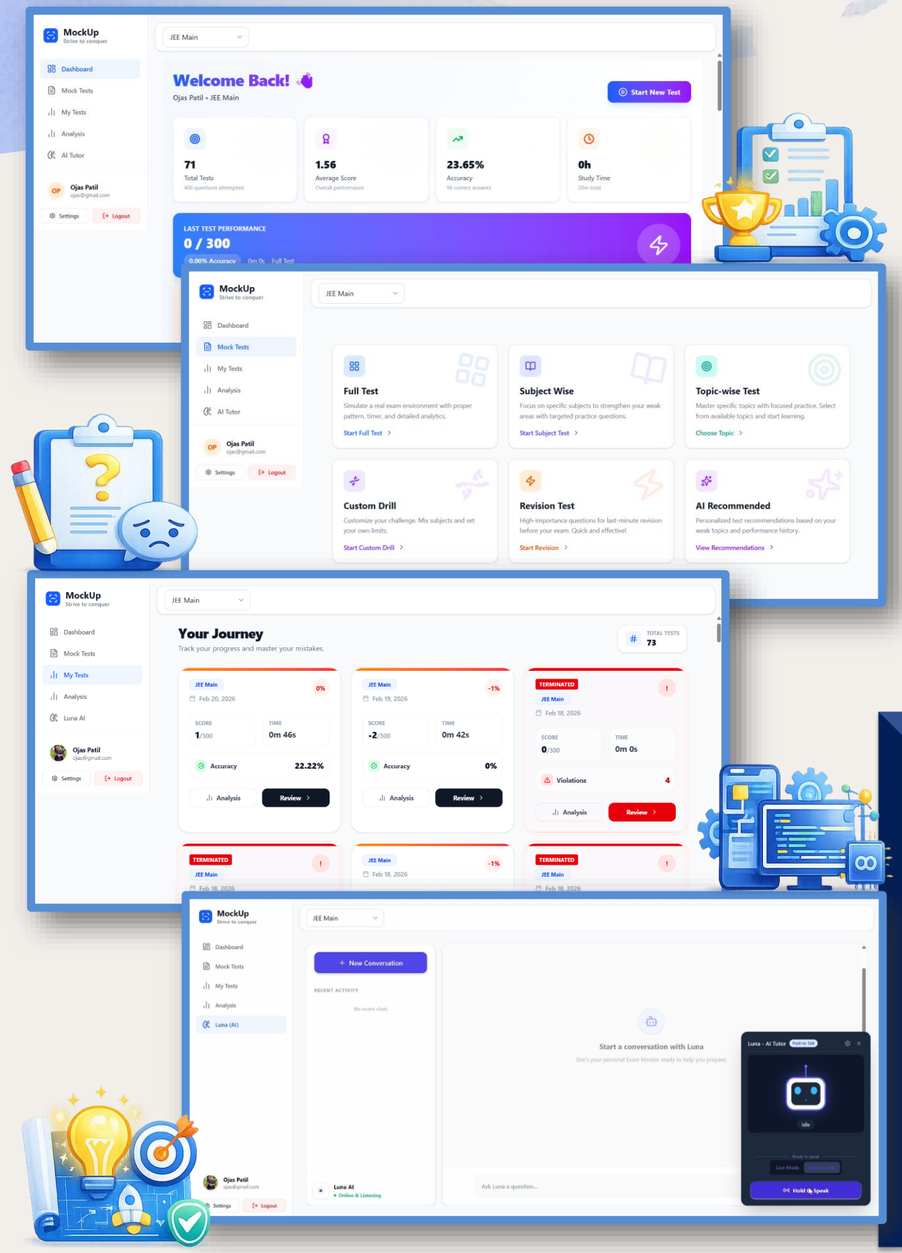
STATE - **MAHARASHTRA**

PROBLEM STATEMENT

Competitive exam aspirants generate large amounts of test and mock exam data, yet most platforms limit analysis to scores and ranks, failing to reveal deeper issues such as conceptual gaps, poor time management, ineffective question selection, or inconsistent performance patterns. Students lack clear, personalized guidance on what to study, how to improve, and where they are losing marks, while static and generic feedback does not adapt to their progress over time. This results in inefficient study plans, wasted effort, and slow improvement, highlighting the need for an AI-powered analytics platform that transforms raw performance data into actionable, adaptive, and personalized insights for effective exam preparation.

PROPOSED SOLUTION

- **MockUp Ai** is an **AI-powered learning and productivity platform** designed to bridge the gap between raw exam data and effective preparation.
- It transforms mock test and exam performance data into **personalized, actionable insights** that help students **study smarter, improve faster, and build exam confidence.**
- By combining intelligent analytics, adaptive learning paths, and real-time exam simulation, the platform functions as a **virtual mentor** that continuously analyzes a **student's strengths, weaknesses, learning behavior, time management, and focus levels**.
- As more data is generated, the system evolves—refining **recommendations**, predicting **performance trends**, and **optimizing study** effort to maximize score improvement.



FEATURES

- Ai-powered Question-level Performance Analysis
- Ai Based Strength and Weakness Identification
- Personalized Intelligent Study Planner With Recommendations
- AI Learning Assistant For Personalized Development
- Ai-based Proctoring & Anti-cheating Detection
- Predictive Exam Score and "What-if" Analysis
- Progress Tracking Dashboards With Charts And Heatmaps
- Exam Anxiety Monitor

PROCESS FLOW DIAGRAM

User Registration

Mock Test / Exam Attempt

Full Length Tests
Subject Wise and Topic Wise Tests
Ai Recommended Tests
Real-Time Exam Monitoring
Detailed Analysis Report

Data Collection & Preprocessing

Performance Data Aggregation

Behavior Data Aggregation

Data Cleaning and Normalization

AI & ML Analysis Engine

Question-Level Performance Analysis
Topic & Sub-Topic Weakness Detection
Error Pattern Identification
Time Management Analysis
Learning Behavior Analysis

Insight Generation Module

- Strength & Weakness Mapping
- Learning Gap Identification
- Productivity Metrics (Effort vs Score Gain)

Personalized Recommendation Engine

Adaptive Study Plan Generation

Targeted Practice Question Selection

Difficulty Level Adjustment

AI Learning Assistant (Tutor Mode)

Explain Incorrect Answers
Provide Conceptual Explanations
Generate Similar Practice Questions
Answer Student Queries

Student Dashboard

Personalized Insights
Progress Charts & Heatmaps
Study Plan & Recommendations
Exam Readiness Score

INNOVATIONS

DIFFERENTIATION

MockUp AI differentiates itself by:

- Using **AI-driven question-level analysis** instead of only topic-level insights
- Continuously adapting learning paths as new test data is generated
- Focusing on **learning productivity and effort optimization**, not just scores
- Providing **predictive insights** rather than only historical performance
- Simulating **real exam conditions** with integrity and focus tracking

PROBLEM SOLVING

- Converts raw mock test scores into **actionable learning insights**
- Analyzes performance at **question, topic, and difficulty level**
- Detects **root causes** of mistakes (concept gaps, time issues, error patterns)
- Generates **personalized and adaptive study plans** for each student
- Recommends **high-impact topics** for maximum score improvement
- Predicts future exam performance based on learning trends

Unique Selling Proposition

AI-Powered Question-Level Intelligence

Predictive Performance Analytics

Real-time mock exam simulation with proctoring

Adaptive Learning & Continuous Improvement

Topic and sub-topic strength & weakness identification

AI Learning Assistant for concept explanation and doubt solving

TECHNOLOGIES

FRONTEND

BACKEND

DATABASE

THANKYOU